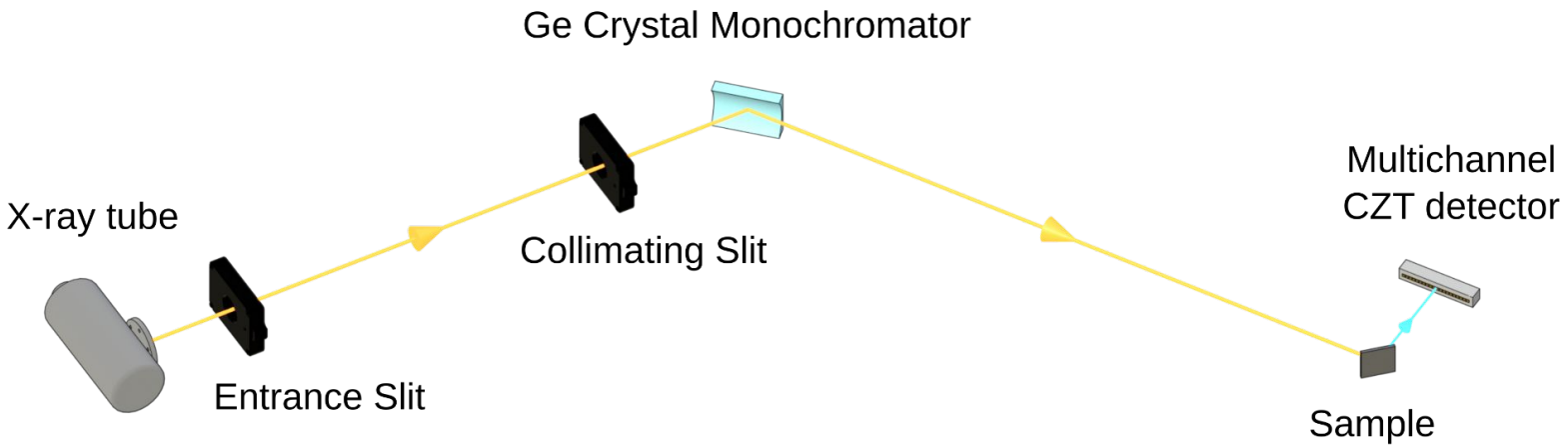
PRISM

Precision Room-temperature Instrumentation for high-energy X-ray absorption Spectroscopy Measurements

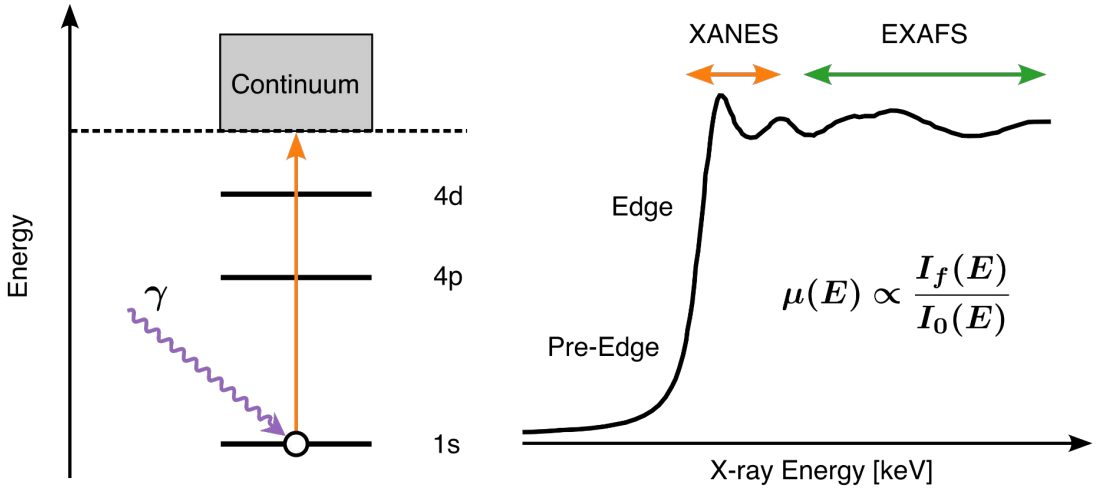
Responsabile di Progetto: Dr. Simone Manti, INFN - LNF
Referente di Progetto: Dr.ssa Catalina Curceanu, INFN - LNF

How can we perform laboratory XAS above 20 keV?



X-ray Absorption Spectroscopy (XAS): chemical state and local structure

XAS probes element-specific **Absorption Edges** to reveal **Chemical State** and the **Local Atomic Environment**.



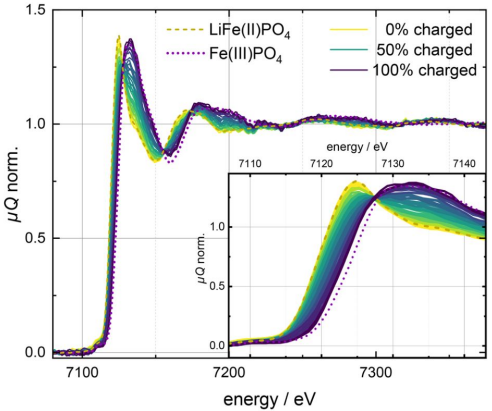
1. **Element selectivity**
   Core-level edges identify the absorbing element
2. **EXAFS**
   Neighbor distances, coordination and disorder
3. **Pre-edge**
   Electronic states and local symmetry
4. **XANES**
   Oxidation state and coordination geometry

State-of-the-Art XAS under operating conditions

Element-specific insight into **Electronic State** and **Local Atomic Structure**.

Battery cycling

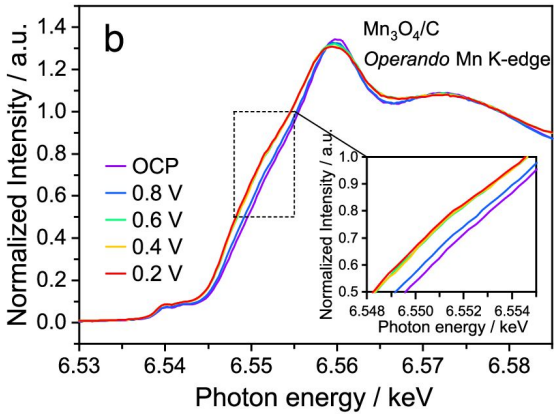
Lab XANES tracks $LiFePO_4$ conversion
Fe K-edge, 15 min per spectrum



Praetz et al., JAAS 40, 2447–2461 (2025), Fig. 2

Working catalysts

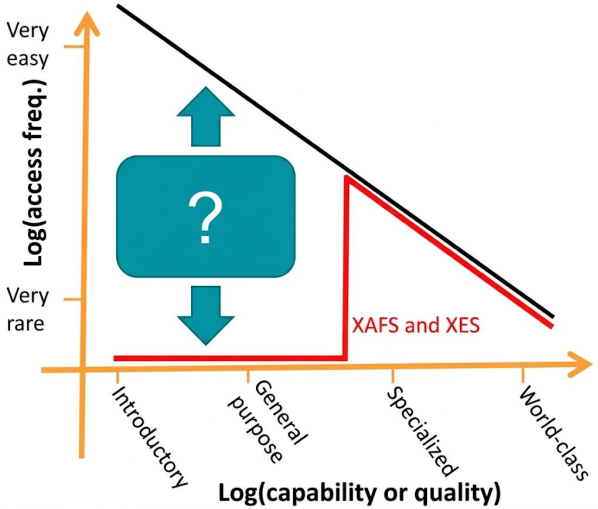
Operando XANES probes the Mn state
under fuel-cell operating conditions



Li et al., Nature Communications (2025), Fig. 3b

Why bring XAS into the laboratory?

Synchrotrons and **FEL** remain the reference for XAS performance.
Laboratory instruments with X-ray tubes offer routine access and flexible measurement times.



J. Seidler, et al. (2024)

- **Routine screening** and **repeated measurements**
- Long **In-Situ / Operando** campaigns
- Sample and protocol optimization for beamtime

Commercial laboratory XAS already exists!

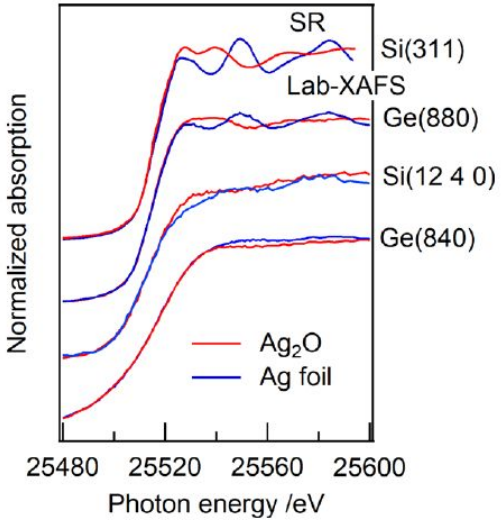


Sigray
≈ 1.5 M€



Laboratory XAS above 20 keV: Opportunities and Challenges

- K-edges of Mo, Pd, Ag and Sn are relevant to **Catalysis** and **Energy Materials**.
- **Closely spaced L-edges** can restrict the accessible EXAFS range.



Feasibility demonstrated in transmission

- Ge(880): ≈12 h for Ag K-edge XANES
- Ag K-edge EXAFS impractical at this count rate

Spectral purity is also critical

- Al filters suppress lower-order reflections (440) and characteristic Ge K lines.

Challenge: quantitative fluorescence XAS with efficient detection and sufficient SNR in a single lab day!

Yamamoto. "Selection of Monochromator Crystal for Pb L- and Ag K-Edge XANES Using a Laboratory-Based Spectrometer." *X-Ray Spectrometry* (2026).

PRISM

MAIN GOAL:

“To develop and validate a room-temperature laboratory platform for fluorescence-mode XAS above 20 keV, combining a high-resolution crystal monochromator with a custom multichannel CZT detector “

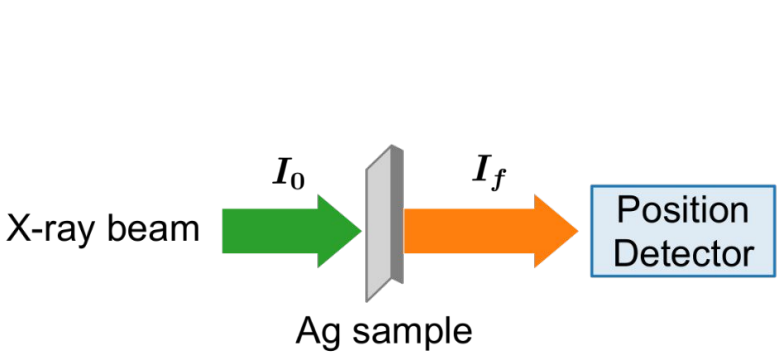
PRISM probes XAS K-edge with the characteristic Fluorescence line

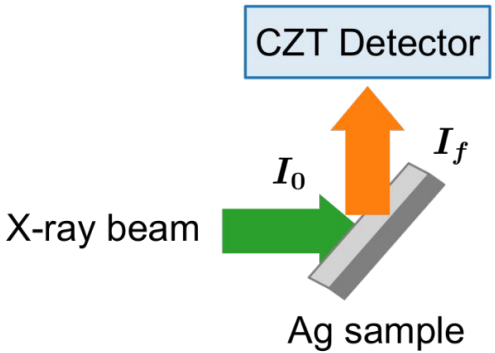
Transmission Mode

Requires sufficient transmission and a uniform sample thickness.

Fluorescence Mode

Enables measurements on dilute samples and samples unsuitable for transmission.

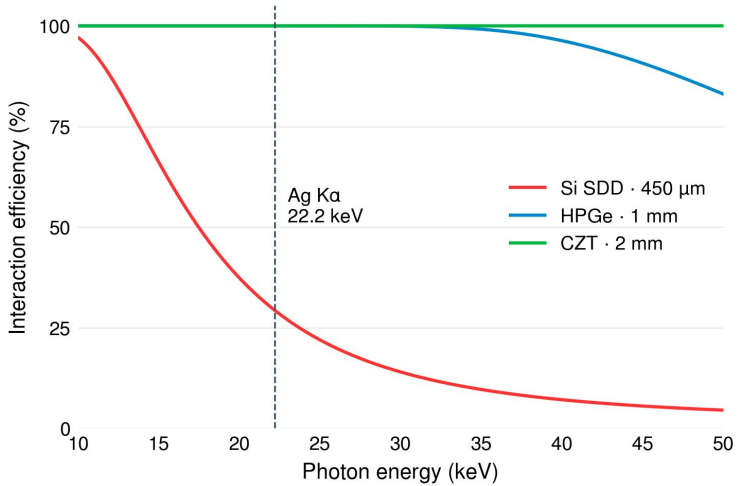




PRISM requires efficient, energy-resolving detection of weak fluorescence above 20 keV.

CZT have High Efficiency > 20 keV with Room-Temperature operation

Fluorescence-Mode XAS above 20 keV requires high detection efficiency



Efficient Fluorescence collection

Higher absorption at Ag Kα (22.2 keV) than **Silicon Drift Detectors (SDDs)**

Spectroscopy at Room-Temperature

No cryogenic cooling, unlike **HPGe detectors**

Quantitative Multichannel Readout

Channel calibration and waveform analysis to address charge sharing and pile-up.

CZT detector, front-end electronics and waveform DAQ ~33% of PRISM's cost

Custom CZT provides access to individual Channels and Waveforms

A detection chain tailored to quantitative fluorescence-mode XAS above 20 keV

Matched sensor and Front-End

20-pixel linear array with an independent low-noise preamplifier for each channel.

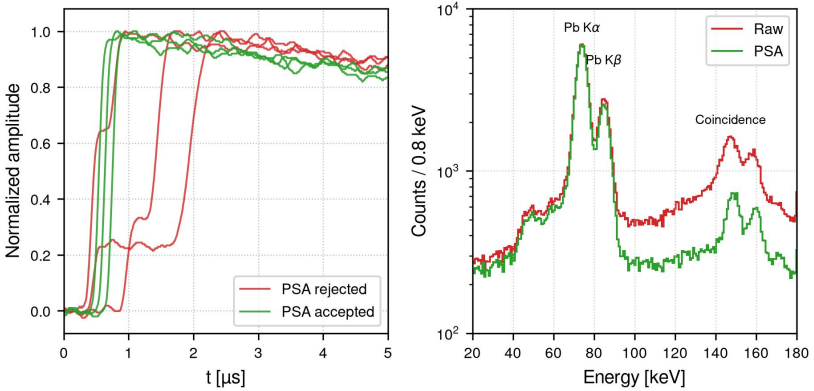
Access to individual Waveforms

Develop charge-sharing reconstruction and identify pile-up and distorted pulses.

Calibration under XAS conditions

Equalize channels and quantify counting losses to extract reliable fluorescence intensities.

Building on prior CZT results at DAΦNE BTF



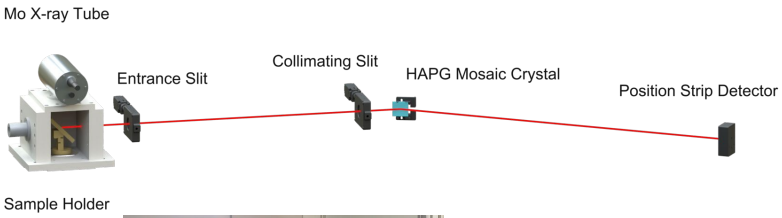
Pulse selection suppresses **Coincidence Structures**

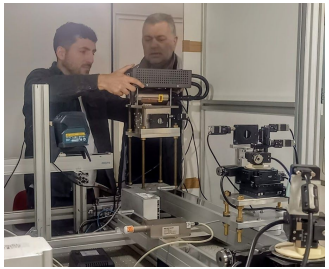
Manti et al., Condensed Matter 11, 38 (2026).

M15 targets: ≤ 2% at Ag Kα, Dead Time < 10%, Pile-up contamination < 5%

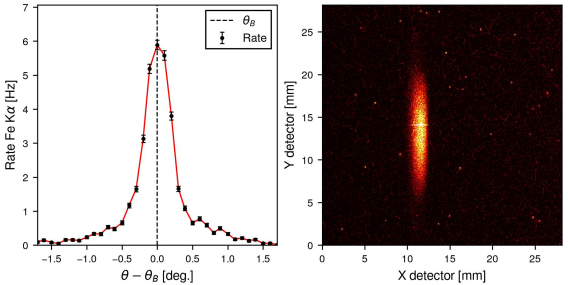
VOXES as the experimental basis for PRISM

Existing VOXES platform

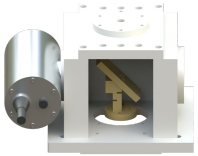




Manti et al., Condensed Matter 9.1 (2024): 19.



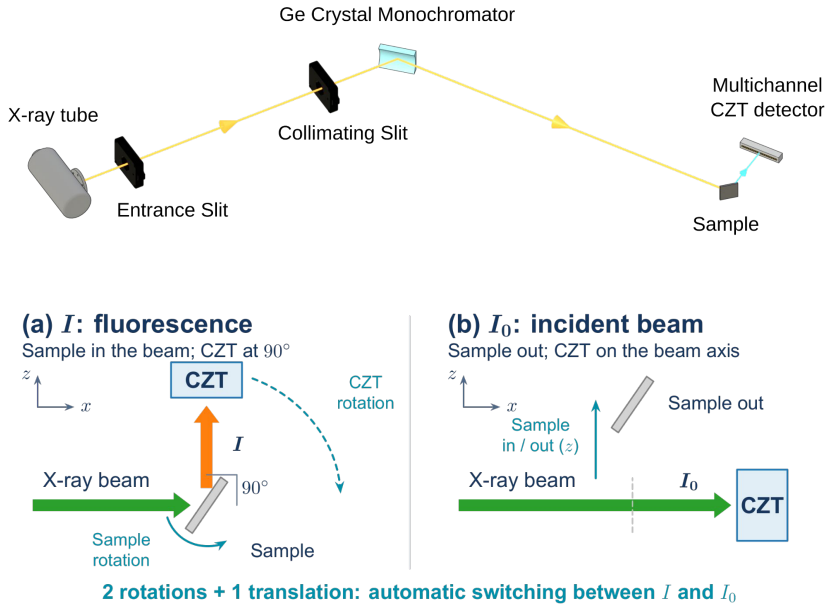
Demonstrated **Angular Scan and Beam Focusing** (e.g. FeKα)



Recent upgrades enable **Absorption measurements**

Manti et al.. Engineering Innovations 19 (2026): 23-29.

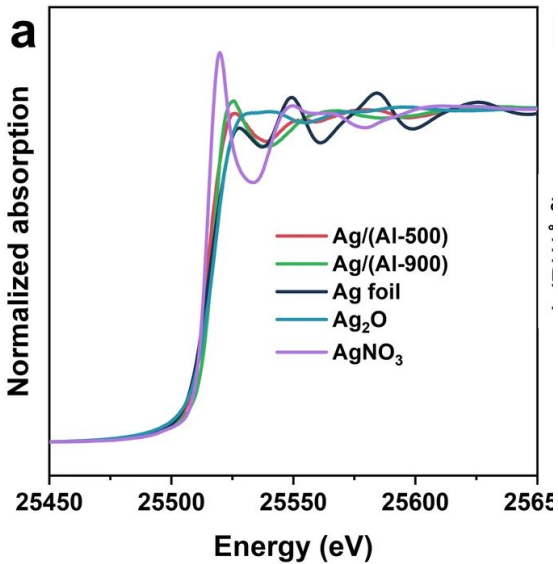
Scanning above 20 keV requires tailored Optics, Source and Motors



- **Ge Crystal · €20k**
  Narrow-band incident-energy selection for XANES: target ≤10 eV FWHM at the Ag K-edge. Ge(880) is the baseline candidate.
- **W-Anode X-ray tube · €17k**
  Higher continuum flux above 20 keV to provide sufficient photons after monochromatization.
- **Motorized Stages · €10k**
  Two rotations and one translation to switch reproducibly between fluorescence I_f and incident-flux I_0 measurements at each energy point.

RAY-TRACING+ GEANT4 End-End simulations will define the final setup specifications

Ag K-edge XANES will be the decisive validation of PRISM



Li, Jiaxin, et al. *Comm.* 15.1 (2024): 3874.

REFERENCE SAMPLES

Ag foil + two Ag(I) compounds

Different local coordination environments

PERFORMANCE TARGETS

≤10 eV FWHM incident-beam bandwidth

≥20 edge-step SNR

VALIDATION

Energy calibration and reproducibility

Comparison with reference XANES spectra

Work Plan and Milestones for PRISM

Work Packages (WP) for Simulation, CZT, Commissioning and Validation and **Milestones (MS)**

WORK PACKAGE (WP)	PLANNED ACTIVITY (A)	2027				2028			
		1-3	4-6	7-9	10-12	13-15	16-18	19-21	22-24
WP1 - Requirements, simulations and design freeze	A1.1 - Requirements definition								
	A1.2 - SHADOW4 ray-tracing optimization								
	A1.3 - GEANT4 fluorescence simulations								
	A1.4 - Final geometry selection			MS1					
WP2 - Room-temperature multichannel CZT spectroscopy for hard X-ray fluorescence XAS	A2.1 - CZT detector and front-end design								
	A2.2 - Multichannel DAQ and signal-processing development								
	A2.3 - CZT calibration and performance characterization								
	A2.4 - Detector validation and operating-point selection					MS2			
WP3 - Instrument integration and commissioning	A3.1 - Procurement of source, optics and motion stages								
	A3.2 - Mechanical design and spectrometer assembly								
	A3.3 - Hardware, software and control-system integration								
	A3.4 - Instrument commissioning and verification						MS3		
WP4 - Ag reference samples and realistic sample demonstration	A4.1 - Ag foil calibration and reproducibility tests								
	A4.2 - XANES validation on Ag reference compounds								
	A4.3 - Fluorescence XAS demonstration on a realistic sample								
	A4.4 - Performance characterization and operating limits								MS4

- **M6 Setup specifications frozen from Simulations**
- **M15 CZT spectroscopic performance demonstrated**
- **M18 Integrated instrument commissioned**
- **M24 Ag XANES demonstration Validated**

Organization and Synergies supporting PRISM

1.7 FTE at LNF, with specialist expertise for CZT, Crystal Optics and Simulations

Defined responsibilities across WP1–WP4

Name	Unit	FTE	Responsibility/WP
Simone Manti	INFN-LNF	1.0	Project Leader; WP1–WP4
Alessandro Scordo	INFN-LNF	0.3	Crystal optics, ray tracing, integration; WP1/WP3
Massimiliano Bazzi	INFN-LNF	0.3	Electronics and DAQ; WP2
Catalina Curceanu	INFN-LNF	0.1	Scientific Supervisor and application validation; WP4
Total		**1.7**	-

VOXES: equipment and test infrastructure.
Palermo / KAONNIS (CSN3): CZT and calibration.
VIP (CSN3): pulse-shape analysis.
Further links: SPHINX and EuAPS/AQUA

ENRICH COST Action
Manti co-leads Working Group 3 with M. Sánchez del Río, principal developer of Ray-Tracing code **SHADOW4**.



Mission at ESRF (Grenoble)

Letter of support for PRISM
Potential technical exchanges on CZT sensors and testing (WP2).

Budget for the integrated PRISM instrument

Item	Year	Cost (k€)
CZT detector and front-end	2027	30
Digitizer and DAQ	2027	23
Bent Ge crystal optics	2028	20
Motorized stages and controllers	2027	10
Cooled W-anode X-ray tube	2028	10
X-ray tube power supply	2028	7
Oral conference travel	2028	4
Ag reference targets	2027	2
Custom mechanical supports	2027	2
DataCloud CPU and storage	2027	2
Additional reference targets	2028	2
Total Requested	**2027–28**	**112**

112 k€

Requested budget
2027: 69 k€
2028: 43 k€

100 k€ in equipment
89% of the request

45 k€

Estimated in-kind value
VOXES tube and bunker
Motion and control systems

Expected impact beyond the PRISM instrument

Fluorescence XAS for catalysis and energy materials above 20 keV

Scientific access	Screen thick or dilute samples and track slow changes. Optimize protocols before synchrotron beamtime.
INFN capabilities	Reuse CZT readout, simulations and calibration. Train staff in hard-X-ray instrumentation.
Technology transfer	Evaluate the modular detector and analysis with industry. Extend to new applications after laboratory validation.

Main Risks and practical Mitigations

Risk	Activities	Mitigation	PNR
R1. Limited accuracy of the end-to-end simulation	A1.2–A1.4	Benchmark against analytical calculations and VOXES data. Consult SHADOW4 developer M. Sánchez del Río through ENRICH WG3.	**4 Medium (2 × 2)**
R2. Extended CZT optimization phase	A2.1–A2.4	Use the VOXES Si PIN for preliminary calibration at reduced efficiency while CZT optimization continues.	**2 Low (1 × 2)**
R3. Delayed delivery of critical components	A3.1–A3.4	Start integration and alignment with compatible VOXES components, including non-optimized crystals.	**4 Medium (2 × 2)**
R4. Lower than expected fluorescence count rate or SNR	A3.3–A4.4	Optimize scan steps, dwell time, acceptance and detector shaping. Preserve energy resolution and background rejection.	**6 High (2 × 3)**

Proposed initial PNR = probability × impact (1–3 each). **Low 1–2** **Medium 3–4** **High 6–9**

PRISM: fluorescence XAS above 20 keV at LNF

24 months to demonstrate quantitative Ag K-edge XANES

Ge crystal optics with custom CZT detection at room temperature

≤10 eV	≥20	112 k€
Target incident-beam FWHM	Target edge-step SNR	Requested budget

VOXES infrastructure, a 1.7 FTE team and a defined validation plan.

References

[1] Yamamoto. "Selection of Monochromator Crystal for Pb L- and Ag K-Edge XANES Using a Laboratory-Based Spectrometer." X-Ray Spectrometry (2026).

[2] Manti, Simone, and Alessandro Scordo. "Recent Developments of the VOXES Von Hamos X-Ray Spectrometer for Laboratory XES and XAS Studies." Engineering Innovations 19 (2026): 23-29.

[3] Scordo, A., et al. "VOXES: a high precision X-ray spectrometer for diffused sources with HAPG crystals in the 2–20 keV range." Journal of Instrumentation 13.04 (2018): C04002-C04002.

[4] Manti, Simone, et al. "Enhancing Performances of the VOXES Bragg Spectrometer for XES Investigations." Condensed Matter 9.1 (2024): 19.

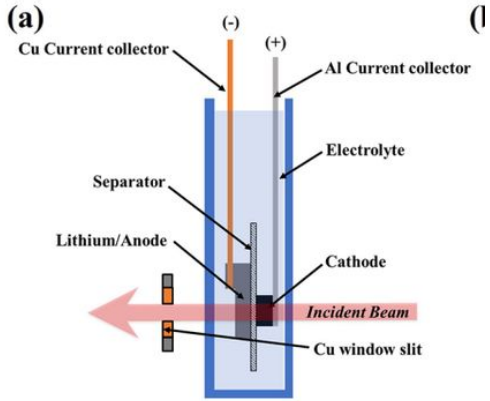
[5] Manti, Simone, et al. "Machine-learning-enhanced event selection for BEGe detectors in the VIP experiment." Journal of Instrumentation 21.01 (2026): P01036.

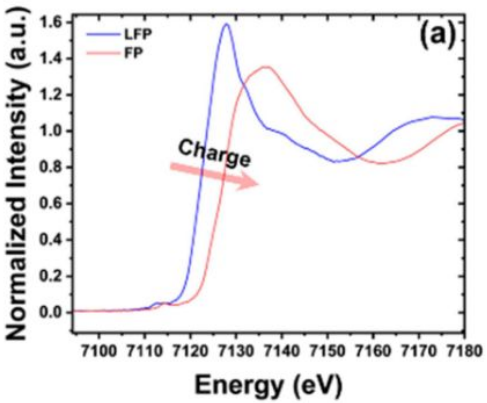
[6] Manti, Simone, et al. "Pulse-Shape Discrimination with Machine Learning for CZT Detectors at DAΦNE Beam Test Facility." Condensed Matter (2026).

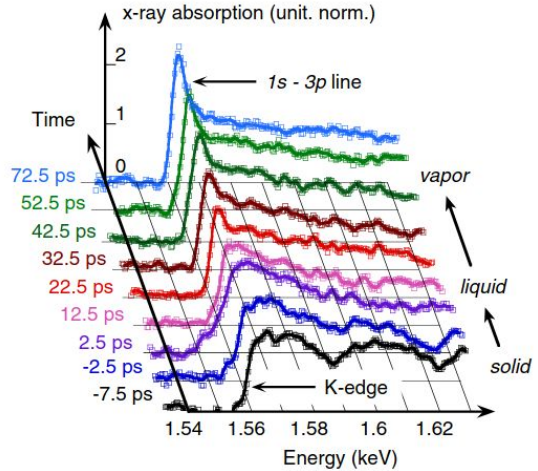
SPARE

State-of-the-Art XAS: tracking materials during operation

- Element-specific sensitivity to **Oxidation State** and **Local Atomic Structure** enables studies of materials under operating conditions.
- **Battery Cycling:** Operando XAS tracks changes in the Fe oxidation state during charging.
- **Phase Transitions:** Ultrafast XANES follows the evolution from solid to liquid and vapor.







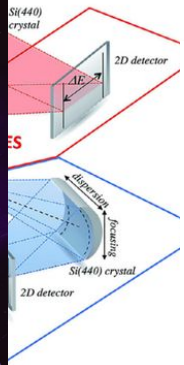
Krishnan et al. *ACS* 16.13 (2024)

Dorchies, F., et al. *PRL* 107.24 (2011)

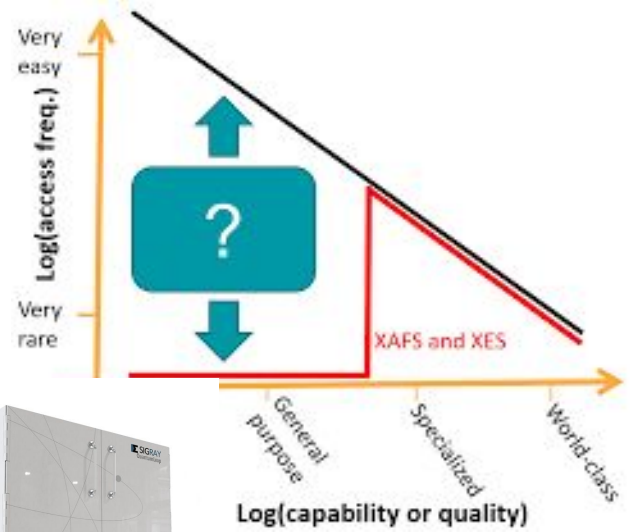



easyXAFS300



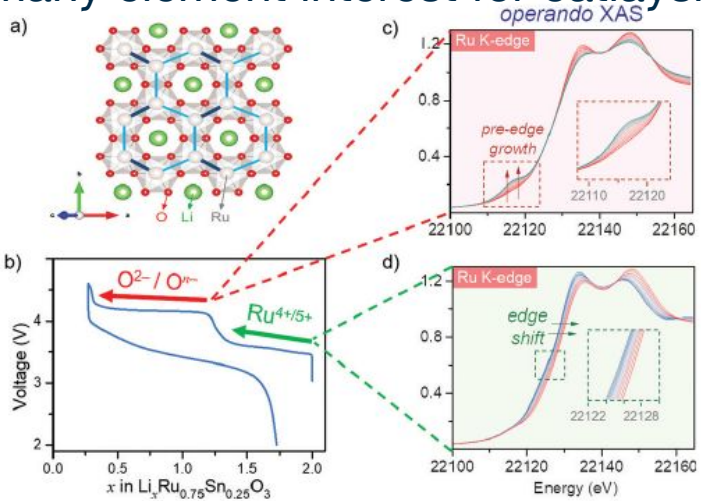

Si(440) crystal
2D detector
ΔE
dispersive
focusing
Si(440) crystal
2D detector

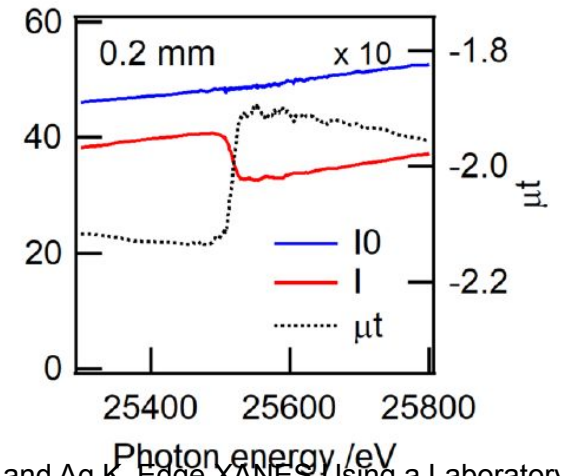



Very easy
Log(access freq.)
Very rare
?
XAFS and XES
General purpose
Specialized
World-class
Log(capability or quality)

Laboratory XAS above 20 keV: opportunities and challenges

- Current limitations are performing XAS for edge above 20 keV, where many element interest for catlaysi (Mo,Pd,Ag,Sn)



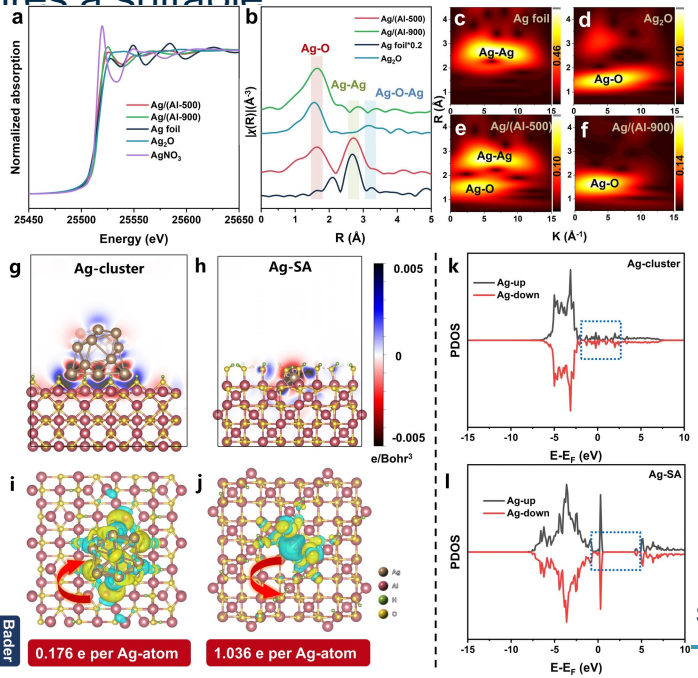


Yamamoto, Takashi. "Selection of Monochromator Crystal for Pb L-, Br K-, and Ag K-Edge XANES Using a Laboratory-Based Spectrometer." *X-Ray Spectrometry* (2026).

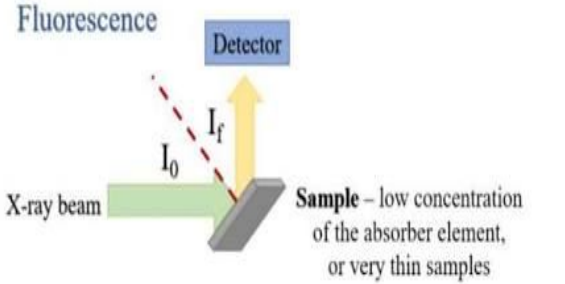
Why fluorescence detection mode for PRISM?

Transmission requires a suitable
homogeneous sar

Fluorescence exte
dilute or constrain







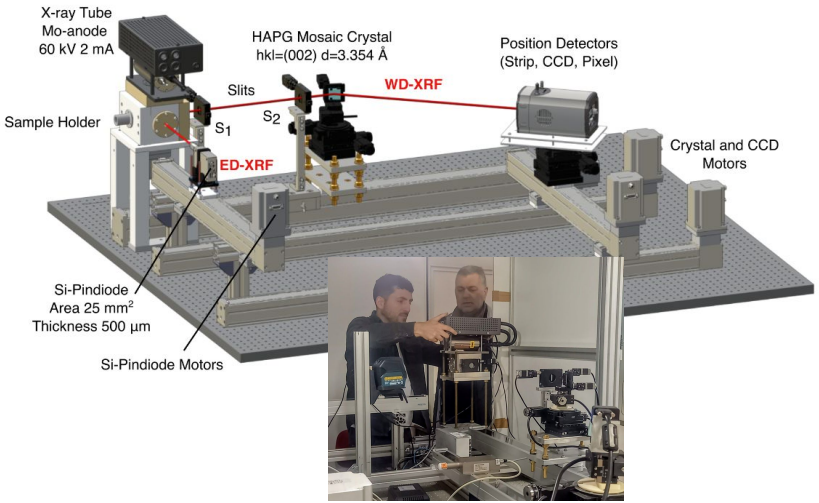
Scan the incident e
Record fluorescenc

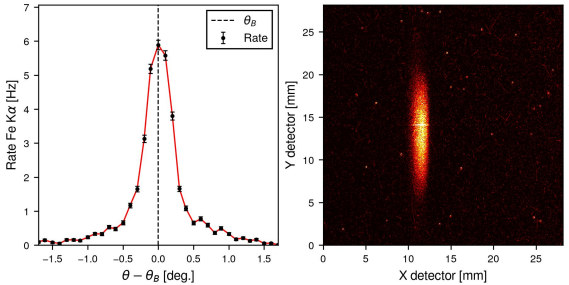
Manti, Simone, and Alessandro
Engineering Innovations 19 (202

Spectrometer for Laboratory XES and XAS Studies."

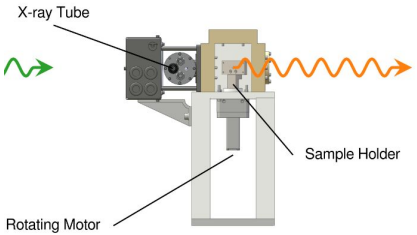
VOXES as the experimental basis for PRISM

Existing instrument and infrastructure





Angular scan and **focused beam** measured with the crystal optics.



Recent upgrades enable **Absorption measurements**

Manti, Simone, et al. "Enhancing Performances of the VOXES Bragg Spectrometer for XES Investigations." Condensed Matter 9.1 (2024): 19.
Manti, Simone, et al.. "Recent Developments of the VOXES Von Hamos X-Ray Spectrometer for Laboratory XES and XAS Studies." Engineering Innovations 19 (2026): 23-29.